



PatchFormer: A novel patch-based transformer for accurate remaining useful life prediction of lithium-ion batteries

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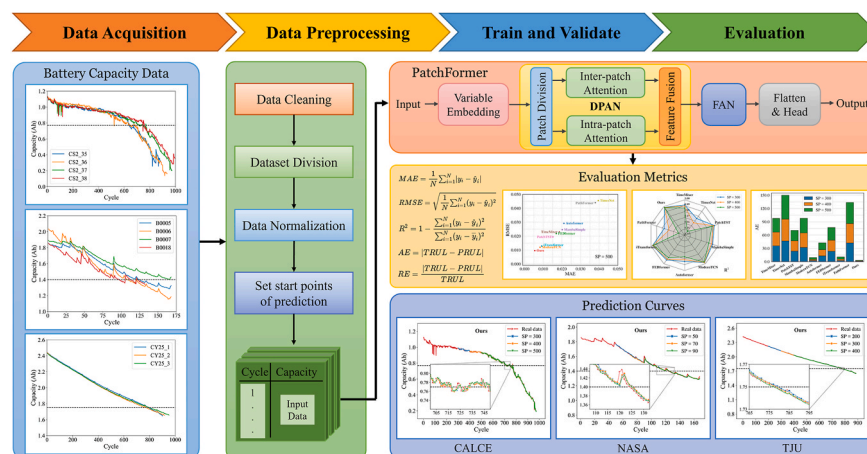
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HIGHLIGHTS

- A novel patch-based Transformer (PatchFormer) is proposed to improve the accuracy of remaining useful life (RUL) prediction.
- PatchFormer combines dual patch-wise attention (DPA) to effectively integrate global correlations and local dependencies.
- The DPA encompassing inter- and intra-patch attention to adaptively fuse and select features within and between patches.
- PatchFormer outperforms various forecasting models in RUL prediction on three public battery degradation datasets.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Remaining useful life (RUL)
Lithium-ion batteries
Machine learning for energy storage
Battery degradation
Time series forecasting (TSF)
Deep learning

ABSTRACT

Accurate prediction of the remaining useful life (RUL) of lithium-ion batteries is critical for ensuring their safe and reliable operation. Nevertheless, achieving precise RUL prediction presents significant challenges due to the intricate degradation mechanisms inherent in battery systems and the influence of operational noise, particularly the capacity regeneration phenomena. To address these issues, we propose a novel patch-based transformer (PatchFormer). The proposed architecture incorporates a Dual Patch-wise Attention Network (DPAN), which effectively captures global correlations between patches via inter-patch attention while also addressing local dependencies within individual patches through intra-patch attention. This dual attention mechanism facilitates the selection and integration of both global and local features. Additionally, we implement a Feature-wise Attention Network (FAN) that utilizes self-attention on the fused temporal features to elucidate the interrelationships among these features. This model proficiently integrates global associations and local details within the partitioned time series, enabling it to accurately delineate the capacity degradation trends while precisely capturing the capacity regeneration occurrences. Extensive experiments are conducted on three public

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<https://doi.org/10.1016/j.jpowsour.2025.236187>

Received 16 November 2024; Received in revised form 26 December 2024; Accepted 5 January 2025

Available online 17 January 2025

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battery degradation datasets, benchmarking our model against state-of-the-art (SOTA) time series forecasting (TSF) models. The results consistently show that our model achieves superior prediction performance across different prediction starting points. The source code and datasets are available at <https://github.com/USTC-AI4EEE/PatchFormer>.

1. Introduction

In recent years, severe energy crises and excessive carbon emissions have become pressing challenges for humanity. Consequently, the demand for clean energy and energy storage has rapidly increased [1–3]. Lithium-ion batteries, as a storage device, are widely used in energy storage systems and electric vehicles due to their broad operational temperature range, high energy density, and environmentally friendly characteristics [4,5]. However, with increased usage, lithium-ion batteries age, leading to capacity degradation, which can potentially cause machinery malfunctions and even catastrophic accidents [6,7]. On the other hand, premature battery replacement can result in resource waste [8,9]. The remaining useful life (RUL) is a key indicator for assessing the health status of lithium-ion batteries. Accurately predicting the RUL of lithium-ion batteries can effectively guide battery replacement strategies, preventing resource waste while mitigating serious accidents caused by battery aging [10].

According to different modeling approaches, lithium-ion battery RUL prediction methods can be roughly divided into: model-based methods and data-driven methods [11,12]. Model-based methods consider battery load conditions, material performance, and degradation mechanisms to predict the RUL, including electrochemical models, equivalent circuit models, and empirical degradation models [13]. However, model-based methods are complex and can be significantly affected by external factors, making it difficult to establish stable models, leading to challenges in their broader practical application.

Data-driven methods leverage vast amounts of real measurement data to learn the nonlinear mapping relationship between relevant parameters and the battery RUL, facilitating precise RUL prediction. These methods offer simple and efficient modeling without the need to account for the complex electrochemical reactions and failure mechanisms within the battery [14]. Overall, data-driven approaches can be divided into traditional machine learning methods and deep learning methods [15]. Common traditional machine learning techniques include Support Vector Machines (SVM) [16,17], Support Vector Regression (SVR) [18,19], and Gaussian Process Regression (GPR) [20,21]. While these methods are simple and efficient for modeling, their ability to perform nonlinear mapping is limited, leading to a gradual replacement by deep learning methods for RUL prediction tasks.

Deep learning methods have emerged as a research hotspot in the field of lithium-ion battery RUL prediction due to their superb feature extraction and nonlinear mapping capabilities [22–24]. Given that the RUL of lithium-ion batteries is strongly correlated with usage time, time series modeling has become central to existing deep learning-based RUL prediction methods [14]. Zhang et al. [25] leveraged the excellent time series forecasting capabilities of Long Short-Term Memory (LSTM) networks to learn long-term dependencies between battery capacities, predicting future aging trajectories. Li et al. [26] connected one-dimensional Convolutional Neural Network (CNN) with Activity State Tracking LSTM (AST-LSTM) to enhance prediction capabilities by capturing hierarchical features and temporal dependencies among multiple variables affecting battery degradation. Zhu et al. [27] combined CNN, bidirectional LSTM (BiLSTM), and attention mechanisms (AM) to improve the accuracy of battery RUL estimates and their long-term estimation performance. Li et al. [28] proposed a novel hybrid model based on Temporal Convolutional Network (TCN) and LSTM for RUL prediction, where TCN extracts local temporal features from the capacity data during degradation, and LSTM further captures global features from the TCN-convolved results. Li et al. [13] used TCN with a

feature attention mechanism to form an encoder module, then combined it with a GRU and temporal attention mechanism to create a decoder module, ultimately outputting prediction results with a Deep Neural Network (DNN). This hybrid model effectively captures the regeneration phenomenon of battery capacity and represents the capacity degradation trend. Wang et al. [22] implemented a Multiple Feature Extraction (MFE) module to extract various characteristics during the charging and discharging cycles, then used a Gated Residual Unit (GRU) module to capture long-term dependencies in the time series data, employing a Time Convolutional Attention (TCA) module to focus on relevant temporal states for more accurate prediction.

In the last few years, some researchers have conducted research specifically on time series forecasting (TSF) problems and proposed a series of models, such as Transformer [29], Autoformer [30], and FEDformer [31], etc., which have promoted the development of TSF research. Transformer-based models have been preliminarily applied to RUL prediction tasks, yielding favorable prediction performance. Chen et al. [32] combined Denoising Autoencoders (DAE) with transformer, integrating denoising and forecasting into a unified framework to achieve superior predictive performance. Han et al. [33] designed a novel denoising transformer-based neural network (DTNN) that utilizes historical data to better forecast the capacity of lithium-ion batteries. Cai et al. [34] proposed a hybrid prediction framework based on complete ensemble empirical mode decomposition with adaptive noise (CEEM-DAN), transformer, and DNN that can accurately predict RUL with minimal capacity data. Table 1 provides a comprehensive summary of numerous studies on the RUL prediction of lithium-ion batteries, offering an overview of the types of input variables, classification, and prediction methods employed.

Although extensive research has been conducted on the RUL prediction of lithium-ion batteries, the accuracy of existing prediction methods is often inadequate due to abrupt changes and severe fluctuations in the capacity degradation curves caused by local capacity regeneration phenomena [35]. In order to further improve the prediction accuracy of RUL of lithium-ion batteries, a patch-based transformer (PatchFormer) is proposed. The model mainly consists of Dual Patch-wise Attention Network (DPAN) and Feature-wise Attention Network (FAN), which can effectively extract and fuse inter-patch and intra-patch features and model the correlation between features. Inspired by the PathFormer [36], the designed DPAN captures global correlations between patches through inter-patch attention, while intra-patch attention focuses on local details within individual patches. It effectively selects and integrates features extracted from both global and local perspectives. Additionally, the constructed FAN utilizes self-attention on the fused temporal features to capture correlations among features. PatchFormer not only accurately represents the degradation trends in battery capacity sequences but also precisely captures the capacity regeneration phenomena in lithium-ion batteries, resulting in higher precision in RUL prediction. The main contributions of this paper can be summarized as follows:

- (1) The proposed PatchFormer combines dual patch-wise attention and time series modeling, effectively integrating the global correlations and local dependencies of the partitioned time series, enabling it not only to accurately represent the degradation trend of the battery capacity sequence, but also to accurately capture the capacity regeneration phenomenon of lithium-ion batteries, thereby improving the accuracy of RUL prediction.

Table 1

The summary of selected studies on RUL prediction.

Classification	Author and year	Reference	Input variables	Prediction Methods
Model-based methods	Yan et al. (2019)	[37]	Capacity data obtained by state-space estimation	Extended Kalman filter (EKF)
	Li et al. (2019)	[38]	Capacity data obtained by filter algorithm	Unscented Kalman filter (UKF)
	Cong et al. (2020)	[39]	Capacity data obtained by state-space estimation	Unscented particle filter (UPF)
Traditional machine learning methods	Kong et al. (2021)	[40]	Voltage-dependent health features and a thermal-dependent health feature	Gaussian process regression (GPR)
	Guo et al. (2022)	[41]	Historical capacity	Relevance vector machine (RVM)
	Hong et al. (2023)	[42]	Characteristic voltage (CV), cycle times, and capacity	A novel particle filtering based recursive least squares (PF-RLS) algorithm
Deep learning methods	Zhang et al. (2018)	[25]	Historical capacity	Long short-term memory recurrent neural network (LSTM-RNN)
	Xu et al. (2022)	[43]	Three hybrid features that combines both electrochemical states and measurement data	LSTM-based encoder-decoder
	Li et al. (2022)	[26]	Historical capacity	CNN-ASTLSTM
	Zhu et al. (2022)	[27]	Capacity and four health indicators (HIs)	Attention-based CNN-BiLSTM
	Li et al. (2023)	[28]	Historical capacity	TCN-LSTM
	Li et al. (2023)	[13]	Historical capacity	TCN-GRU-DNN
	Wang et al. (2024)	[22]	Within-cycle features and overall features from the cycles	MFE-GRU-TCA
	Chen et al. (2022)	[32]	Historical capacity	Combined DAE and transformer network (DeTransformer)
	Han et al. (2023)	[33]	Historical capacity	Denoising transformer-based neural network (DTNN)
	Cai et al. (2023)	[34]	Historical capacity	CEEMDAN-transformer-DNN
	Chen et al. (2024)	[44]	Historical capacity	Self-attention with Mixture of Experts (AttMoE)

- (2) The designed DPAN leverages a dual attention mechanism to adaptively integrate inter-patch and intra-patch features, taking into account the abruptness and volatility of battery capacity degradation curves. By employing this dual attention approach, which specifically targets both inter-patch and intra-patch dynamics, the DPAN effectively captures the global dependencies and local characteristics of time series data, thereby significantly enhancing the accuracy of RUL predictions.
- (3) Compared with various SOTA TSF models, the proposed PatchFormer consistently shows superior performance in RUL

prediction on three public battery degradation datasets. In addition, the proposed model is less affected by the prediction starting point and has excellent generalization ability.

The subsequent sections of this study are structured as follows: Section 2 elucidates the datasets used, data preprocessing, and dataset division. Section 3 delves into the details and experimental procedures of the proposed method. Section 4 introduces the model's hyperparameter settings, evaluation metrics, comparisons with various state-of-the-art TSF models on different datasets and ablation studies of the proposed model. Finally, Section 5 summarizes the key conclusions drawn from this study.

2. Datasets

2.1. Dataset description

To verify the performance of the proposed model, three public datasets available from Center for Advanced Life Cycle Engineering (CALCE dataset) [45], NASA Prognostics Center of Excellence (NASA dataset) [46] and Tongji University (TJU dataset) [47] are used in this paper.

2.1.1. CALCE dataset

The battery cells of CALCE dataset are prismatic batteries with LiCoO₂ cathode and have a rated capacity of 1.1 Ah. In our study, we select four batteries numbered CS2_35, CS2_36, CS2_37 and CS2_38. All batteries were repeated charging and discharging process. They were charged at 0.55 A constant current until the battery voltage reached 4.2 V and were continued at 4.2 V constant voltage until the charge current was lower than 50 mA. These lithium-ion batteries had the same discharge cut-off voltage of 2.7 V and the same discharge constant current of 1.1 A, and the whole process was completed at 1 °C. It is generally accepted that a battery is considered to have reached its end of life (EOL) when its actual useable capacity decreases to 70 %–80 % [14] of its rated capacity. The CALCE dataset defines 70 % of its rated capacity as EOL, which is taken as 0.77 Ah in this paper.

2.1.2. NASA dataset

NASA dataset consists of 18650-typed lithium cobalt oxide (LiCoO₂) batteries with a rated capacity of 2 Ah. The charge-discharge protocol of the NASA dataset is similar to that of the CALCE dataset. During battery charging process, the lithium-ion batteries were subjected to a constant current of 1.5 A at room temperature until the battery voltage reached 4.2 V and were continued at 4.2 V constant voltage until the current dropped to 20 mA. Discharge was carried out at a constant current level of 2 A until the battery voltage fell to 2.7 V, 2.5 V, 2.2 V and 2.5 V for B0005, B0006, B0007, and B0018, respectively. Generally, the NASA dataset defines 70 % of its rated capacity as EOL, which is taken as 1.40 Ah.

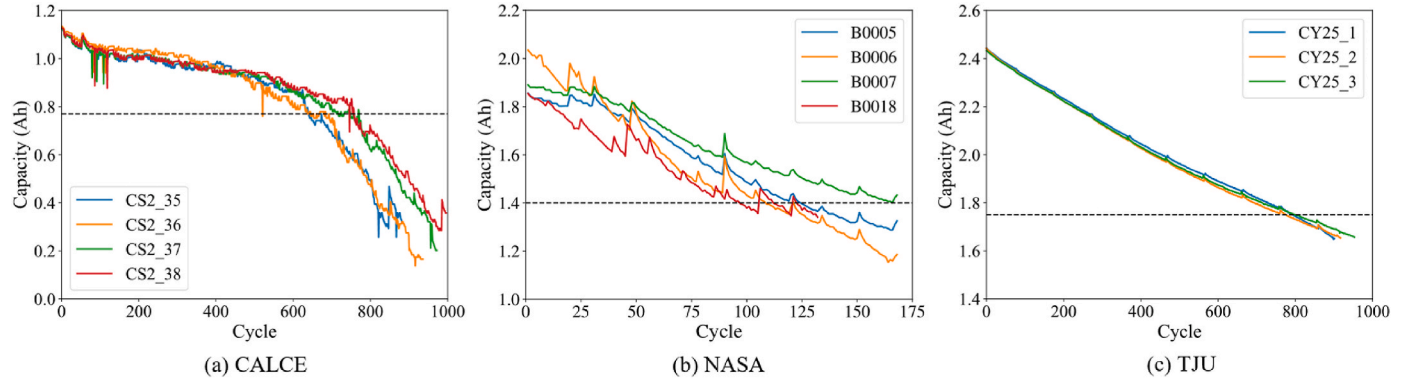
2.1.3. TJU dataset

TJU dataset consists of batteries with 42 (3) wt.% Li(NiCoMn)O₂ blended with 58 (3) wt.% Li(NiCoAl)O₂ positive electrode (NCM + NCA battery), which have a rated capacity of 2.5 Ah. NCM batteries and NCA batteries are currently mainstream lithium-ion batteries and are widely used in many fields such as new energy vehicles. In this study, we select a group of batteries with a charge and discharge current rate of 0.5C/1C, named CY25_1, CY25_2, and CY25_3. The voltage and current data include the charge, discharge, and relaxation processes. The cell cycling was performed by constant current charging at a current rate of 0.5C (1.25A) to 4.2V, followed by a constant voltage charging step at 4.2V until a current of 0.05C (125 mA) was reached. The NCM + NCA cells were then discharged to 2.5V at a current rate of 1C (2.5A). The relaxation time between constant voltage charging and constant current discharging for the NCM + NCA battery is 60 min, and the sampling time

Table 2

Batteries information of CALCE, NASA and TJU datasets.

Source	Battery ID	Charge/Discharge Cut-off Voltage	Charge/Discharge Constant Current	Temperature	Rated Capacity	Failure Threshold
CALCE	CS2-35	4.2V/2.7V	0.55A/1.1A	1 °C	1.1Ah	0.77Ah
	CS2-36	4.2V/2.7V	0.55A/1.1A	1 °C	1.1Ah	0.77Ah
	CS2-37	4.2V/2.7V	0.55A/1.1A	1 °C	1.1Ah	0.77Ah
	CS2-38	4.2V/2.7V	0.55A/1.1A	1 °C	1.1Ah	0.77Ah
NASA	B0005	4.2V/2.7V	1.5A/2A	24 °C	2Ah	1.4Ah
	B0006	4.2V/2.5V	1.5A/2A	24 °C	2Ah	1.4Ah
	B0007	4.2V/2.2V	1.5A/2A	24 °C	2Ah	1.4Ah
	B0018	4.2V/2.5V	1.5A/2A	24 °C	2Ah	1.4Ah
TJU	CY25-1	4.2V/2.5V	1.25A/2.5A	25 °C	2.5Ah	1.75Ah
	CY25-2	4.2V/2.5V	1.25A/2.5A	25 °C	2.5Ah	1.75Ah
	CY25-3	4.2V/2.5V	1.25A/2.5A	25 °C	2.5Ah	1.75Ah

**Fig. 1.** Capacity degradation curves for lithium-ion batteries in three datasets: (a) CALCE dataset, (b) NASA dataset, (c) TJU dataset.**Table 3**

Dataset division of CALCE I, CALCE II, NASA and TJU datasets.

Datasets	Training data	Test data
CALCE I	CS2_36 CS2_37 CS2_38	CS2_35
CALCE II	CS2_35 CS2_36 CS2_38	CS2_37
NASA	B0006 B0007 B0018	B0005
TJU	CY25_2 CY25_3	CY25_1

is 30 s. TJU dataset defines 70 % of its rated capacity as EOL, which is 1.75 Ah. Table 2 provides detailed information about the lithium-ion batteries used in this study.

2.2. Data preprocessing

In the CALCE dataset, each cycle consists of multiple steps, with each step corresponding to an index (represented as Step_Index). To obtain the discharge capacity, data points with Step_Index = 7 (representing the discharge process) must be filtered out from each cycle. The discharge capacity is then calculated using the coulombic integral method. The NASA dataset comprises three main cycles: charging, discharging, and impedance. The discharge capacity for each battery can be obtained from the discharge cycles, revealing the overall aging trend. The basic data in the TJU dataset includes charging, discharging, and relaxation processes. The discharge capacity for each cycle can be extracted from the discharge process, allowing for the determination of the battery's capacity degradation trajectory.

The Pauta criterion (also known as the 3 σ criterion) assumes that a set of measurement data contains only random errors, calculating the standard deviation and establishing a confidence interval based on a certain probability. Any errors that fall outside this interval are considered outliers and should be removed, as they do not represent random errors. In the CALCE and TJU datasets, which include a large number of charging and discharging cycles, measurement errors may be

introduced, leading to outliers when calculating cumulative discharge capacity. In such cases, the Pauta criterion is applied because it assumes that the data distribution possesses the following characteristics:

$$\begin{cases} P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.6826 \\ P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.9544 \\ P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.9974 \end{cases} \quad (1)$$

where $P(\cdot)$ represents the probability that the data falls within the corresponding confidence interval, μ denotes the mean of the data, and σ indicates the corresponding standard deviation.

It can be assumed that these values are nearly entirely concentrated within the interval $[\mu - 3\sigma, \mu + 3\sigma]$, with the probability of falling outside this range being less than 0.3 %. In this study, as shown in Eq. (1), we apply a two-fold standard deviation interval to remove outliers from the cumulative discharge capacity obtained from the CALCE dataset, and a three-fold standard deviation interval to filter out outliers from the cumulative discharge capacity obtained from the TJU dataset. The processed capacity degradation curves of three datasets are illustrated in Fig. 1.

To ensure that the training data is on the same scale and to minimize the impact of distribution variations on the neural network, it is essential to normalize the input data prior to training. In this regard, this paper employs the min-max normalization method to scale the values to the range of [0, 1]. This method can be expressed as follows:

$$x_{scale} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (2)$$

where x denotes the original battery capacity value, and x_{scale} is the normalized value.

2.3. Dataset division

Following the partitioning of the CALCE dataset presented in Ref. [27], battery CS2_35 and battery CS2_37 are selected as testing

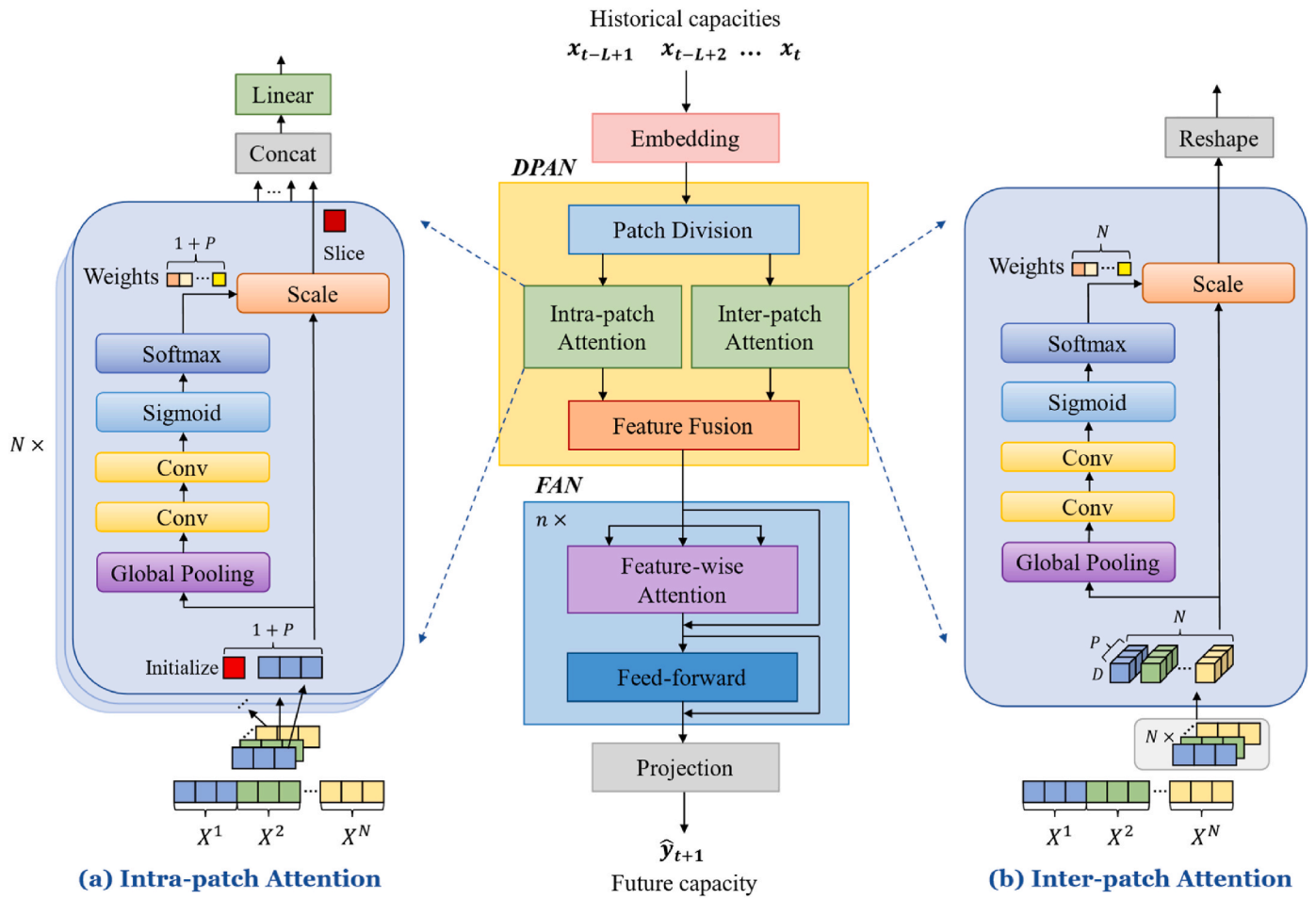


Fig. 2. The overall architecture of PatchFormer. (a) Intra-patch Attention. (b) Inter-patch Attention.

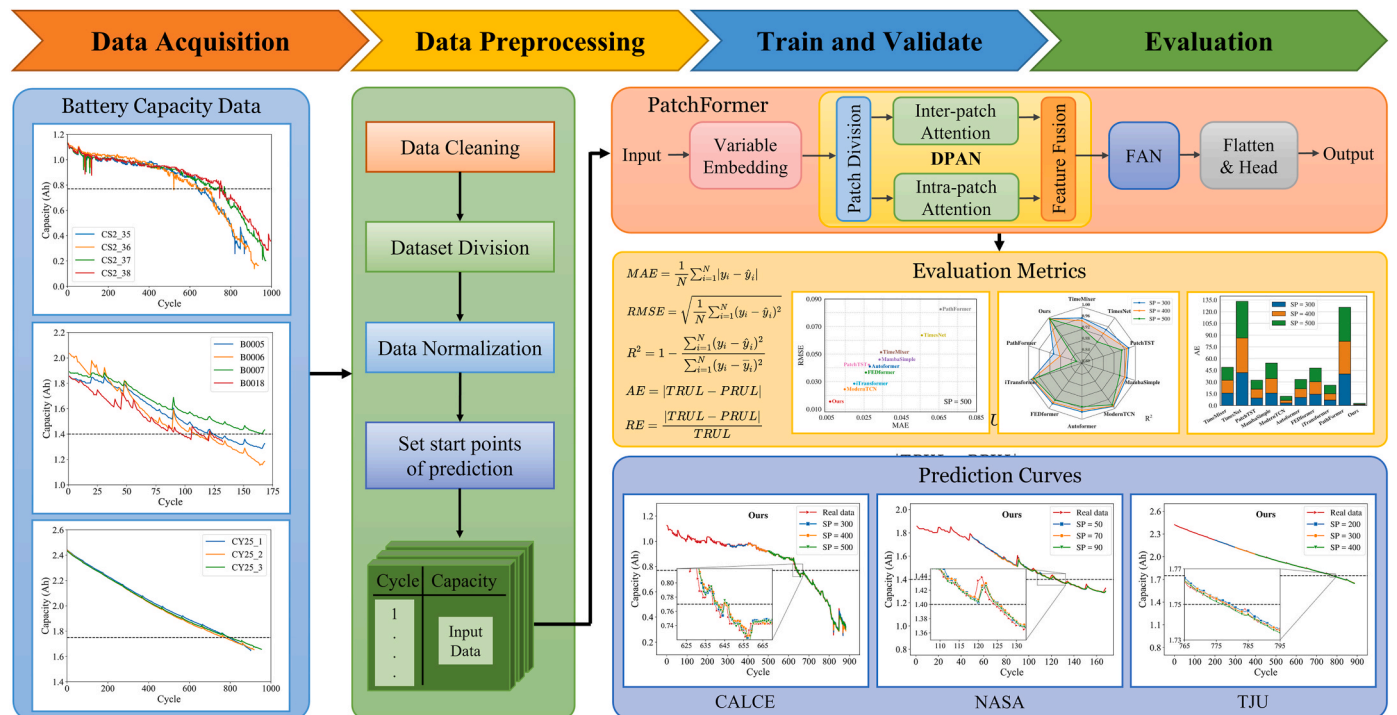


Fig. 3. Experimental flow of the proposed method.

Table 4

The hyperparameter optimization and settings of the PatchFormer model for CALCE I, CALCE II, NASA and TJU datasets.

	Hyperparameters	CALCE I/ II	NASA	TJU
Optimization	Patch length	{2, 4, 8}	{2, 3, 5}	{2, 4, 8}
	Number of units in DPAN	[16, 256]	[16, 256]	[8, 256]
	Factor of FAN	[1,5]	[1,5]	[1,5]
	Number of attention heads of FAN	[1,8]	[1,8]	[1,8]
	Number of hidden layers of FAN	[1,4]	[1,4]	[1,4]
	Gradient clip value	[0.1, 0.5]	[0.1, 0.5]	[0.1, 0.5]
	Dropout rate	[0.01, 0.2]	[0.01, 0.2]	[0.01, 0.2]
Settings	Learning rate	[1e-4, 1e-2]	[1e-4, 1e-2]	[1e-4, 1e-2]
	Sequence length	64	30	64
	Batch size	128	16	128
	Number of units in FAN	64	30	64

batteries to form the CALCE I and CALCE II datasets respectively, while the remaining batteries are used for training. For the NASA dataset, batteries B0006, B0007, and B0018 are used for training, and battery B0005 is used for testing. For the TJU dataset, batteries CY25_2 and CY25_3 are used for training, and battery CY25_1 is used for testing. The specific dataset division is shown in Table 3. The dataset is divided into a training dataset and a testing dataset. The first 80 % of the training dataset is used for training, and the last 20 % is used for validation. For the testing dataset, the model is evaluated from the respective prediction starting points.

3. Method

The overall architecture of the proposed PatchFormer model is illustrated in Fig. 2. This model consists of four main components: an embedding layer at the input end, Dual Patch Attention Network

(DPAN), Feature Attention Network (FAN), and a projection layer at the output end. In this study, historical capacities are used as the model's input. The embedding layer maps the input variable into a high-dimensional space, facilitating subsequent feature extraction. The designed DPAN captures global correlations between patches through inter-patch attention and local dependencies within individual patches through intra-patch attention, enabling the adaptive extraction and effective integration of global relationships and local details in the partitioned temporal series. The FAN utilizes self-attention on the merged features to capture the correlations between them and employs feedforward network (FFN) to model the dependencies within the time series. Finally, the projection layer outputs the capacity prediction results.

3.1. Variable embedding layer

In lithium-ion battery RUL prediction, it is common to use capacity as the input variable of the prediction model. This paper uses a linear mapping layer to map a single variable to a D -dimensional embedding vector to facilitate subsequent modeling of the time and feature dimensions:

$$X_{emb} = \text{Embedding}(X_{in}) \quad (3)$$

where $X_{in} \in \mathbb{R}^{L \times M}$ represents the input of the model, and $X_{emb} \in \mathbb{R}^{L \times D}$ represents the corresponding embedding vector.

3.2. Dual patch-wise Attention Network (DPAN)

To address the sudden changes and significant fluctuations in the capacity degradation curve of lithium-ion batteries caused by local capacity regeneration phenomena, it is crucial to extract local detail features from the time series in addition to global features. Therefore, this study proposes a DPAN to achieve effective feature extraction and enhance the predictive performance of the model. As shown in Fig. 2, the core structure of this network consists of three parts: patch division, dual

Table 5

Evaluation metrics of different models for CALCE I dataset. SP: starting point of prediction; TRUL: true value of RUL; PRUL: predicted value of RUL.

Dataset	Method	SP	TRUL	PRUL	MAE	RMSE	R ²	AE	RE	AMAE	ARMSE	AR ²	AAE	ARE
CALCE I	TimeMixer	300	341	356.8	0.0244	0.0400	0.9608	15.8	0.0465	0.0291	0.0451	0.9455	16.3	0.0780
		400	241	257.4	0.0287	0.0441	0.9512	16.4	0.0683					
		500	141	157.7	0.0341	0.0513	0.9245	16.7	0.1193					
	TimesNet	300	341	383.0	0.0369	0.0482	0.9469	42.0	0.1235	0.0461	0.0555	0.9237	44.4	0.2146
		400	241	285.3	0.0456	0.0547	0.9304	44.3	0.1846					
		500	141	188.0	0.0559	0.0636	0.8938	47.0	0.3357					
	PatchTST	300	341	350.5	0.0200	0.0320	0.9750	9.5	0.0279	0.0235	0.0366	0.9643	10.8	0.0525
		400	241	252.4	0.0230	0.0357	0.9683	11.4	0.0475					
		500	141	152.5	0.0276	0.0422	0.9497	11.5	0.0821					
	MambaSimple	300	341	356.8	0.0239	0.0364	0.9682	15.8	0.0465	0.0284	0.0409	0.9568	18.1	0.0887
		400	241	259.6	0.0281	0.0402	0.9607	18.6	0.0775					
		500	141	160.9	0.0333	0.0460	0.9416	19.9	0.1421					
	ModernTCN	300	341	344.0	0.0109	0.0195	0.9909	3.2	0.0094	0.0127	0.0218	0.9878	3.9	0.0204
		400	241	244.3	0.0124	0.0215	0.9888	3.3	0.0138					
		500	141	146.3	0.0147	0.0244	0.9838	5.3	0.0379					
	Autoformer	300	341	351.1	0.0216	0.0336	0.9738	10.3	0.0303	0.0246	0.0371	0.9656	11.1	0.0539
		400	241	251.8	0.0242	0.0366	0.9681	11.0	0.0458					
		500	141	152.6	0.0281	0.0410	0.9550	12.0	0.0857					
	FEDformer	300	341	355.2	0.0197	0.0301	0.9782	14.4	0.0424	0.0227	0.0332	0.9715	16.0	0.0781
		400	241	256.7	0.0224	0.0328	0.9735	15.9	0.0662					
		500	141	158.0	0.0260	0.0366	0.9628	17.6	0.1257					
	iTransformer	300	341	347.9	0.0141	0.0229	0.9880	6.9	0.0203	0.0168	0.0256	0.9839	8.6	0.0440
		400	241	249.1	0.0166	0.0254	0.9849	8.1	0.0337					
		500	141	151.9	0.0198	0.0285	0.9787	10.9	0.0779					
	PathFormer	300	341	381.3	0.0431	0.0640	0.9060	40.3	0.1185	0.0541	0.0726	0.8695	41.9	0.2010
		400	241	282.9	0.0532	0.0713	0.8814	41.9	0.1746					
		500	141	184.4	0.0659	0.0825	0.8212	43.4	0.3100					
	Ours	300	341	340.4	0.0057	0.0129	0.9962	0.8	0.0023	0.0063	0.0141	0.9951	0.9	0.0045
		400	241	240.2	0.0062	0.0140	0.9954	0.8	0.0034					
		500	141	141.1	0.0069	0.0155	0.9937	1.1	0.0078					

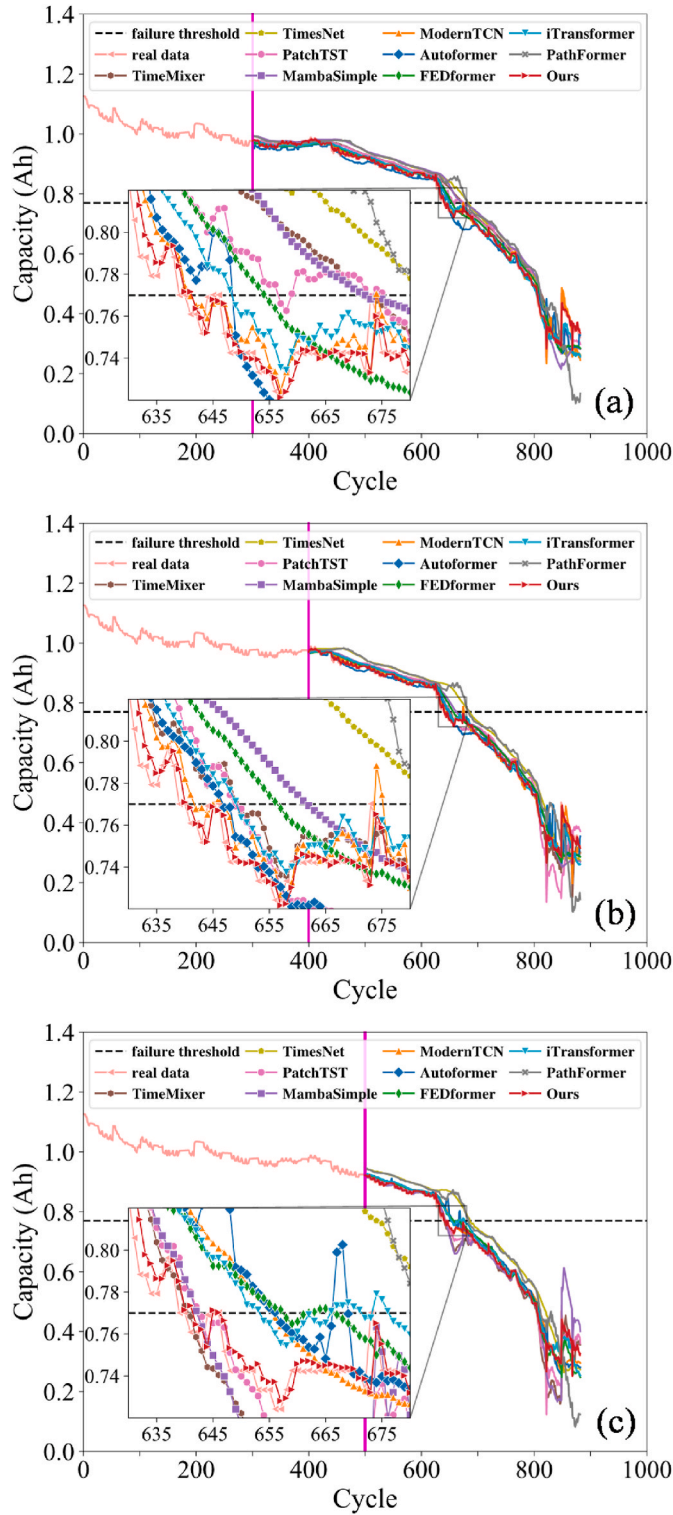


Fig. 4. Comparison of RUL prediction curves from different methods for CALCE I dataset. (a) prediction curves for SP = 300; (b) prediction curves for SP = 400; (c) prediction curves for SP = 500.

patch-wise attention, and feature fusion. The patch division part segments each time series based on a smaller patch length. The dual patch-wise attention part extracts features from between and within patches to capture the global correlations and local dependencies of the partitioned time series. The feature fusion part efficiently integrates the global and local features from the partitioned time series to produce the final time feature output.

3.2.1. Patch division

After variable embedding, the time sequence is denoted as $\mathbf{X} \in \mathbb{R}^{L \times D}$, where L represents the length of the time series, D represents the dimension of the features, and each patch division operation with the patch size P divides $\mathbf{X}$ into N (where $N = L/P$) patches as $(\mathbf{X}^1, \mathbf{X}^2, \dots, \mathbf{X}^N)$, where each patch $\mathbf{X}^i \in \mathbb{R}^{P \times D}$ contains P time steps.

3.2.2. Dual patch-wise attention

Based on the patch division, a dual attention mechanism is proposed to model the temporal dependencies within the divided patches. The dual attention mechanism consists of two parts: intra-patch attention (modeling the dependencies within each individual patch) and inter-patch attention (capturing the dependencies between different patches). As illustrated in Fig. 2(a) and (b), both attention mechanisms are implemented based on Convolutional Squeeze-and-Excitation Attention (CSE-Attention), which is introduced below.

3.2.2.1. CSE-attention. This paper proposes a new attention structure named CSE-Attention for extracting global and local features of partitioned time series. The structure is based on the Squeeze-and-Excitation mechanism [48], which sequentially adopts global average pooling, two consecutive convolutional layers, Sigmoid activation, Softmax normalization and scaling. The main operations of CSE-Attention include Squeeze, Excitation, and Scale.

Taking inter-patch attention as an example, the Squeeze operation employs global average pooling across patches to directly compress the feature map of $D \times P \times N$ containing global information into a $1 \times 1 \times N$ feature vector, that is, to turn each two-dimensional patch into a single value with a global receptive field, shielding the spatial distribution information, and making better use of the correlation between patches. The Squeeze operation is defined as follows:

$$Z = F_{sq}(U) = \text{GlobalPooling}(U) \quad (4)$$

where $U \in \mathbb{R}^{N \times D \times P}$ denotes the input to the CSE-Attention, $Z \in \mathbb{R}^N$ represents the output after the Squeeze operation applied to U , N indicates number of patches, D means the size of feature dimension and P signifies the patch size.

The Excitation operation involves passing statistics of a feature patch through a non-linear bottleneck, which is implemented in two convolutional layers. This allows the model to learn the relative importance of each feature patch. The Excitation operation is defined as follows:

$$S = F_{ex}(Z, W) = \text{Softmax}(\text{Sigmoid}(\text{Conv}_2(\text{Conv}_1(Z)))) \quad (5)$$

where $S \in \mathbb{R}^N$ is the output of $Z \in \mathbb{R}^N$ after the Excitation operation. $\text{Conv}_1(\cdot)$ and $\text{Conv}_2(\cdot)$ are both convolutional layers. The former compresses the number of channels, while the latter restores the number of channels. Here, the number of channels is equal to the number of patches. $\text{Sigmoid}(\cdot)$ is a commonly used activation function, which maps the element values of the excited vector Z to between 0 and 1. $\text{Softmax}(\cdot)$ normalizes the weight vector after activation so that the sum of the weights of each patch is 1.

Finally, the Scale operation applies the learned scale to the original feature map through element multiplication. This will recalibrate the feature map based on the importance of what they have learned. The Scale operation is defined as follows:

$$\tilde{U} = F_{scale}(U, S) = U \odot \text{Expand}(S, \text{shape}(U)) \quad (6)$$

where $\tilde{U} \in \mathbb{R}^{N \times D \times P}$ denotes the output of the CSE-Attention, and the symbol $\odot$ represents element-wise matrix multiplication. The $\text{Expand}(\cdot)$ operation denotes the process of expanding the normalized attention weight S to match the size of U .

3.2.2.2. Intra-patch attention. Consider a set of patches $(\mathbf{X}^1, \mathbf{X}^2, \dots, \mathbf{X}^N)$ divided by patch size P , and the intra-patch attention establishes a

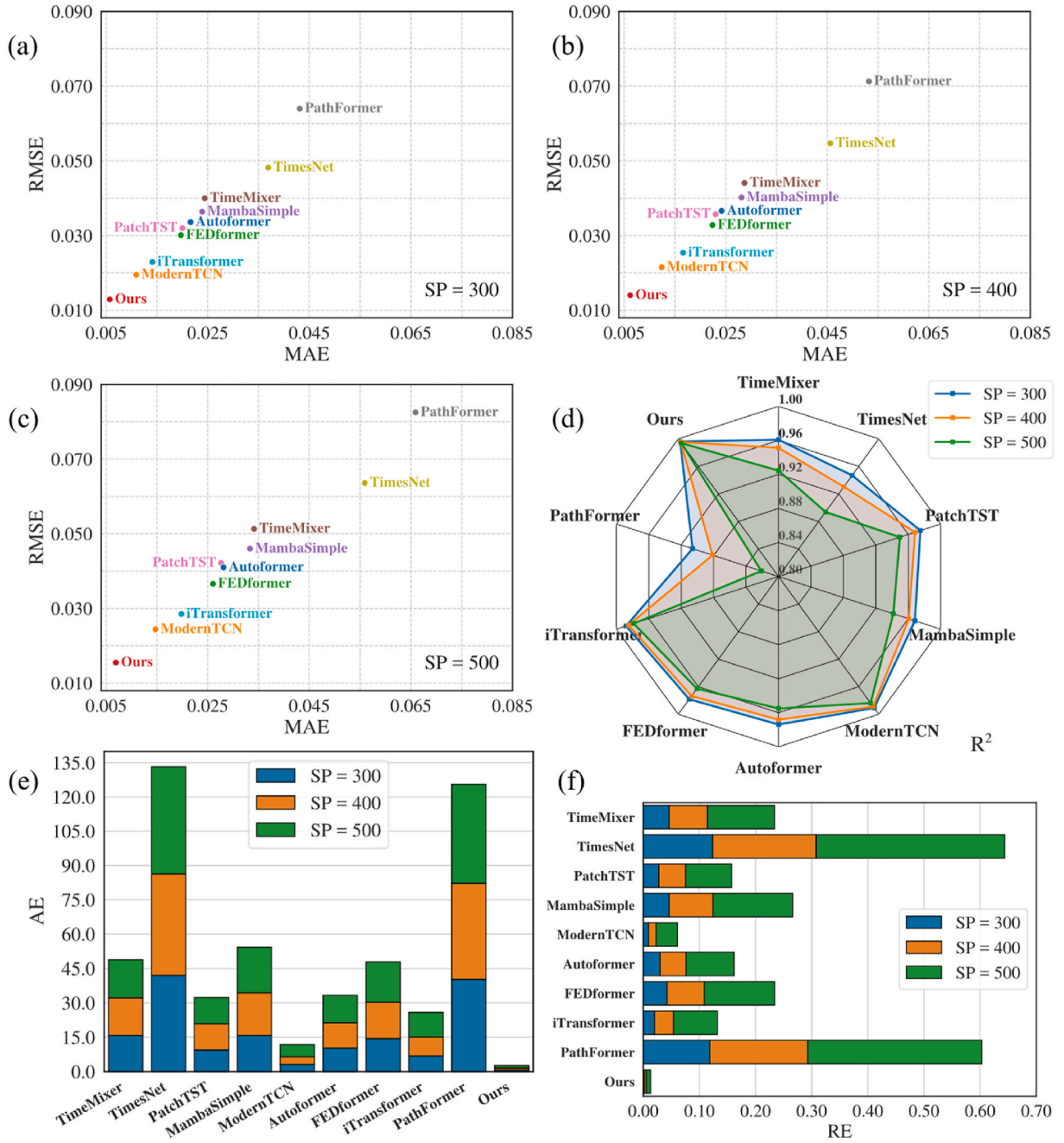


Fig. 5. Evaluation metrics of different methods for CALCE I dataset. (a) MAE and RMSE for SP = 300. (b) MAE and RMSE for SP = 400. (c) MAE and RMSE for SP = 500. (d) R^2 . (e) AE. (f) RE.

relationship between the time steps within each patch. As shown in Fig. 2(a), for the i -th patch $X^i \in \mathbb{R}^{P \times D}$, we first generate a specific embedding vector for the patch through an embedding generator, and concatenate the embedding vector and the patch in the channel dimension (patch size) to obtain a concatenated patch $X_{intra}^i \in \mathbb{R}^{(1+P) \times D}$. Then, X_{intra}^i is passed to the CSE-Attention to assign different weights to different time steps in the concatenated patch. The output of the CSE-Attention is then sliced to obtain the corresponding embedding vector, which can extract local details within the patch X^i :

$$Attn_{intra}^i = \text{Slice}(\text{CSEAttention}(X_{intra}^i)) \quad (7)$$

After CSE-Attention and slicing, the length of each patch is changed from the original input length P to 1. The attention results from all patches are concatenated to produce the output of intra-patch attention

on the divided patches as $Attn_{intra}^i \in \mathbb{R}^{N \times D}$, which represents the local details of nearby time steps in the time series:

$$Attn_{intra} = \text{Concat}(Attn_{intra}^1, Attn_{intra}^2, \dots, Attn_{intra}^N) \quad (8)$$

Finally, the concatenated results are further processed by a linear layer and dimension reordering to obtain the final temporal local features $X_{local} \in \mathbb{R}^{L \times D}$.

3.2.2.3. Inter-patch attention. Inter-patch attention establishes relationships between patches to capture global correlations. As illustrated in Fig. 2(b), for the patch-divided time series $X_{inter} \in \mathbb{R}^{N \times P \times D}$, CSE-Attention is used to model the correlation between patches and obtain the output $Attn_{inter} \in \mathbb{R}^{N \times P \times D}$, which involves the interaction between

Table 6

Evaluation metrics of different models for CALCE II dataset.

Dataset	Method	SP	TRUL	PRUL	MAE	RMSE	R ²	AE	RE	AMAE	ARMSE	AR ²	AAE	ARE
CALCE II	TimeMixer	300	417	452.8	0.0145	0.0188	0.9913	35.8	0.0861	0.0163	0.0207	0.9889	32.6	0.1093
		400	317	347.8	0.0159	0.0204	0.9895	30.8	0.0975					
		500	217	248.2	0.0186	0.0230	0.9860	31.2	0.1444					
	TimesNet	300	417	463.6	0.0267	0.0326	0.9749	46.6	0.1120	0.0325	0.0386	0.9630	50.0	0.1729
		400	317	366.2	0.0312	0.0378	0.9659	49.2	0.1557					
		500	217	271.2	0.0395	0.0455	0.9481	54.2	0.2509					
	PatchTST	300	417	440.7	0.0121	0.0163	0.9931	23.7	0.0570	0.0135	0.0177	0.9912	23.6	0.0801
		400	317	340.4	0.0132	0.0175	0.9917	23.4	0.0741					
		500	217	240.6	0.0152	0.0194	0.9889	23.6	0.1093					
	MambaSimple	300	417	449.1	0.0151	0.0197	0.9902	32.1	0.0772	0.0174	0.0220	0.9872	32.7	0.1115
		400	317	349.6	0.0170	0.0216	0.9879	32.6	0.1032					
		500	217	250.3	0.0202	0.0246	0.9834	33.3	0.1542					
	ModernTCN	300	417	419.4	0.0067	0.0100	0.9976	2.4	0.0058	0.0074	0.0109	0.9971	3.1	0.0115
		400	317	319.6	0.0073	0.0107	0.9973	2.6	0.0082					
		500	217	221.4	0.0083	0.0120	0.9964	4.4	0.0204					
	Autoformer	300	417	423.3	0.0168	0.0245	0.9846	13.1	0.0315	0.0188	0.0267	0.9811	14.2	0.0492
		400	317	323.1	0.0184	0.0264	0.9819	13.9	0.0440					
		500	217	222.2	0.0211	0.0292	0.9769	15.6	0.0722					
	FEDformer	300	417	440.4	0.0129	0.0175	0.9923	23.4	0.0563	0.0147	0.0195	0.9900	25.7	0.0894
		400	317	342.1	0.0143	0.0192	0.9906	25.1	0.0794					
		500	217	245.6	0.0170	0.0218	0.9870	28.6	0.1324					
	iTransformer	300	417	419.5	0.0076	0.0109	0.9972	2.5	0.0060	0.0083	0.0118	0.9966	3.6	0.0134
		400	317	320.2	0.0082	0.0116	0.9967	3.2	0.0101					
		500	217	222.2	0.0092	0.0129	0.9958	5.2	0.0241					
	PathFormer	300	417	459.2	0.0256	0.0314	0.9765	42.2	0.1014	0.0312	0.0373	0.9651	44.6	0.1536
		400	317	360.8	0.0300	0.0364	0.9679	43.8	0.1386					
		500	217	264.7	0.0379	0.0440	0.9509	47.7	0.2208					
	Ours	300	417	417.6	0.0047	0.0083	0.9984	1.0	0.0024	0.0051	0.0089	0.9981	1.0	0.0034
		400	317	317.8	0.0050	0.0088	0.9981	1.0	0.0032					
		500	217	217.8	0.0055	0.0096	0.9977	1.0	0.0046					

Table 7

Evaluation metrics of different models for NASA dataset.

Dataset	Method	SP	TRUL	PRUL	MAE	RMSE	R ²	AE	RE	AMAE	ARMSE	AR ²	AAE	ARE
NASA	TimeMixer	50	75	79.8	0.0239	0.0285	0.9540	8.0	0.1081	0.0228	0.0297	0.8948	7.9	0.1656
		70	55	52.7	0.0241	0.0298	0.9014	6.7	0.1241					
		90	35	26.0	0.0203	0.0307	0.8290	9.0	0.2647					
	TimesNet	50	75	78.0	0.0364	0.0478	0.8753	3.0	0.0405	0.0292	0.0385	0.8600	3.0	0.0608
		70	55	57.9	0.0298	0.0403	0.8349	2.9	0.0537					
		90	35	38.0	0.0213	0.0275	0.8699	3.0	0.0882					
	PatchTST	50	75	79.2	0.0260	0.0319	0.9405	4.2	0.0568	0.0204	0.0260	0.9316	4.4	0.0920
		70	55	56.7	0.0201	0.0250	0.9333	3.9	0.0722					
		90	35	30.4	0.0150	0.0212	0.9209	5.0	0.1471					
	MambaSimple	50	75	81.7	0.0301	0.0362	0.9254	6.7	0.0905	0.0246	0.0304	0.9077	5.5	0.1056
		70	55	60.7	0.0250	0.0305	0.9034	5.7	0.1056					
		90	35	32.9	0.0188	0.0244	0.8943	4.1	0.1206					
	ModernTCN	50	75	74.9	0.0166	0.0213	0.9750	2.9	0.0392	0.0130	0.0186	0.9644	4.7	0.1069
		70	55	51.4	0.0127	0.0172	0.9700	4.4	0.0815					
		90	35	28.2	0.0098	0.0173	0.9483	6.8	0.2000					
	Autoformer	50	75	74.8	0.0234	0.0336	0.9341	3.6	0.0486	0.0226	0.0330	0.8738	4.0	0.0827
		70	55	52.3	0.0230	0.0340	0.8721	4.1	0.0759					
		90	35	32.8	0.0213	0.0313	0.8153	4.2	0.1235					
	FEDformer	50	75	74.9	0.0215	0.0266	0.9569	4.7	0.0635	0.0195	0.0255	0.9249	5.2	0.1089
		70	55	53.6	0.0199	0.0262	0.9217	5.0	0.0926					
		90	35	32.6	0.0172	0.0237	0.8961	5.8	0.1706					
	iTransformer	50	75	69.6	0.0076	0.0141	0.9891	5.4	0.0730	0.0080	0.0152	0.9730	4.7	0.0950
		70	55	50.9	0.0078	0.0149	0.9772	4.3	0.0796					
		90	35	30.7	0.0086	0.0165	0.9528	4.5	0.1324					
	PathFormer	50	75	70.3	0.0274	0.0375	0.9228	5.1	0.0689	0.0225	0.0305	0.9090	6.2	0.1350
		70	55	50.0	0.0214	0.0292	0.9100	5.6	0.1037					
		90	35	27.1	0.0186	0.0248	0.8941	7.9	0.2324					
	Ours	50	75	75.1	0.0056	0.0118	0.9924	0.1	0.0014	0.0062	0.0128	0.9807	0.4	0.0088
		70	55	55.4	0.0061	0.0127	0.9835	0.4	0.0074					
		90	35	34.6	0.0068	0.0140	0.9663	0.6	0.0176					

patches and represents the global correlations of the time series:

$$\mathbf{Attn}_{inter} = \mathbf{CSEAttention}(\mathbf{X}_{inter}) \quad (9)$$

Then, the output of the inter-patch attention $\mathbf{Attn}_{inter}$ is re-dimensionalized to obtain the temporal global features $\mathbf{X}_{global} \in \mathbb{R}^{L \times D}$.

3.2.3. Feature fusion

In order to effectively fuse the global correlation and local details captured by dual attention, we first concatenate the output $\mathbf{X}_{local}$ of the intra-patch attention module and the output $\mathbf{X}_{global}$ of the inter-patch attention module along the time dimension. Then, we perform feature

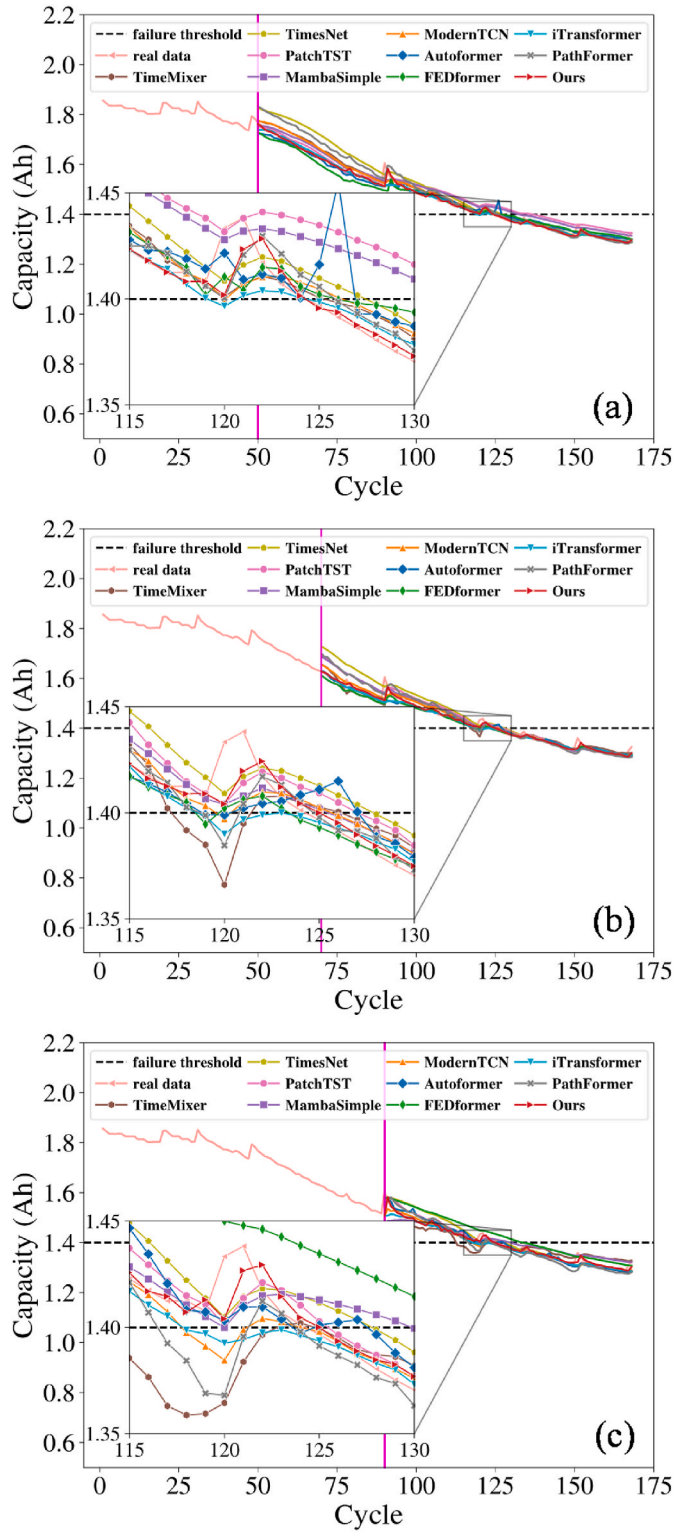


Fig. 6. Comparison of RUL prediction curves from different methods for NASA dataset. (a) prediction curves for SP = 50; (b) prediction curves for SP = 70; (c) prediction curves for SP = 90.

selection and fusion of temporal local features and temporal global features through CSE-Attention:

$$Attn_{fusion} = CSEAttention(Concat(X_{local}, X_{global})) \quad (10)$$

Finally, we separate the output $Attn_{fusion}$ into local features and global

features again, and concatenate them along the feature dimension to obtain the final feature-fused time series $\hat{X} \in \mathbb{R}^{L \times (2D)}$.

3.3. Feature-wise Attention Network (FAN)

After the temporal features are fused, the FAN models the feature dimension. FAN is stacked by multiple transformer sublayers. Each transformer sublayer is divided into two parts: multi-head self-attention and FFN. The multi-head self-attention part uses the attention of the feature dimension to capture the correlation between features. The FFN part learns the interaction of temporal information within each time series.

Given the representation of the $(n-1)$ -th transformer sublayer, H^{n-1} and h parallel attention functions, the i -th ($i \in [1, h]$) attention is defined:

$$head_i = Attention(H^{n-1}W_{Q_i}^n, H^{n-1}W_{K_i}^n, H^{n-1}W_{V_i}^n) \quad (11)$$

where L represents the temporal dimension of the input of the FAN, $W_{Q_i}^n$, $W_{K_i}^n$, and $W_{V_i}^n \in \mathbb{R}^{L \times L_h}$ are projection weights corresponding to the i -th head. Let Q , K , and V denote query, key, and value, scaled dot-product attention defined as follows:

$$Attention(Q, K, V) = softmax(QK^T / \sqrt{L_h})V \quad (12)$$

where $L_h = L/h$, which is employed to avoid generating extremely small gradients and produce a softer attention distribution [29].

Then, the multi-head attention is defined as follows:

$$MultiHead(H^{n-1}) = Concat(head_1, head_2, \dots, head_h)W^O \quad (13)$$

where W^O is a trainable weight.

FFN, which has two different mappings (linear and $ReLU$ nonlinear), is applied to each time step identically and separately. Then, we obtain H^n from the previous $MultiHead(H^{n-1})$ as follows:

$$H^n = FFN(MultiHead(H^{n-1})) \quad (14)$$

$$FFN(x) = ReLU(xW_1 + b_1)W_2 + b_2 \quad (15)$$

3.4. Projection layer

Finally, to predict unknown capacity, a linear layer is used to map the representation learned by the last encoder sub-layer to arrive at the final prediction. The forward propagation process of the projection layer is as follows:

$$\hat{Y} = Head(Flatten(X_{out})) \quad (16)$$

where $X_{out} \in \mathbb{R}^{(2D) \times L}$ is the output of the FAN, $\hat{Y} \in \mathbb{R}^T$ is the final prediction output of PatchFormer and T represents the length of prediction.

3.5. Loss function

In this study, the models are trained by minimizing the validation loss, where the loss function is the Symmetric Mean Absolute Percentage Error (SMAPE). The function of SMAPE is given as follows:

$$SMAPE = \frac{100\%}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \quad (17)$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and N is the number of samples in the validation set. SMAPE has a value between 0 and 2, with smaller values indicating higher prediction accuracy.

Table 8

Evaluation metrics of different models for TJU dataset.

Dataset	Method	SP	TRUL	PRUL	MAE	RMSE	R ²	AE	RE	AMAE	ARMSE	AR ²	AAE	ARE
TJU	TimeMixer	200	579	582.8	0.0071	0.0089	0.9963	7.6	0.0131	0.0093	0.0118	0.9886	6.7	0.0143
		300	479	479.0	0.0086	0.0112	0.9919	6.0	0.0126					
		400	379	374.7	0.0123	0.0152	0.9777	6.5	0.0172					
	TimesNet	200	579	580.0	0.0163	0.0204	0.9830	1.0	0.0017	0.0146	0.0179	0.9807	0.5	0.0011
		300	479	479.2	0.0146	0.0179	0.9808	0.2	0.0004					
		400	379	378.6	0.0130	0.0155	0.9783	0.4	0.0011					
	PatchTST	200	579	587.2	0.0078	0.0092	0.9960	9.8	0.0170	0.0071	0.0088	0.9946	7.7	0.0159
		300	479	484.0	0.0069	0.0084	0.9952	7.4	0.0155					
		400	379	380.4	0.0067	0.0088	0.9925	5.8	0.0153					
	MambaSimple	200	579	588.5	0.0119	0.0145	0.9902	9.5	0.0164	0.0107	0.0129	0.9888	7.6	0.0157
		300	479	486.7	0.0107	0.0129	0.9887	7.7	0.0161					
		400	379	384.3	0.0094	0.0112	0.9874	5.5	0.0146					
	ModernTCN	200	579	581.8	0.0014	0.0022	0.9998	2.8	0.0048	0.0015	0.0023	0.9997	2.7	0.0058
		300	479	481.7	0.0015	0.0023	0.9997	2.7	0.0056					
		400	379	381.7	0.0016	0.0024	0.9995	2.7	0.0071					
	Autoformer	200	579	580.3	0.0050	0.0062	0.9979	5.5	0.0095	0.0049	0.0061	0.9969	5.2	0.0112
		300	479	480.4	0.0049	0.0061	0.9970	5.4	0.0113					
		400	379	380.0	0.0047	0.0060	0.9957	4.8	0.0127					
	FEDformer	200	579	576.2	0.0058	0.0070	0.9977	6.0	0.0104	0.0056	0.0068	0.9966	5.5	0.0118
		300	479	476.5	0.0056	0.0068	0.9968	5.7	0.0119					
		400	379	376.9	0.0054	0.0067	0.9954	4.9	0.0130					
	iTransformer	200	579	580.8	0.0017	0.0023	0.9998	1.8	0.0031	0.0018	0.0025	0.9996	1.7	0.0035
		300	479	480.8	0.0018	0.0025	0.9996	1.8	0.0038					
		400	379	380.4	0.0019	0.0026	0.9994	1.4	0.0037					
	PathFormer	200	579	569.1	0.0100	0.0131	0.9930	9.9	0.0171	0.0121	0.0165	0.9797	12.5	0.0279
		300	479	466.9	0.0112	0.0156	0.9855	12.1	0.0253					
		400	379	363.4	0.0152	0.0209	0.9606	15.6	0.0413					
	Ours	200	579	579.2	0.0013	0.0019	0.9999	0.8	0.0014	0.0014	0.0020	0.9997	0.9	0.0019
		300	479	478.9	0.0014	0.0020	0.9997	0.5	0.0011					
		400	379	378.6	0.0015	0.0022	0.9996	1.3	0.0032					

3.6. RUL prediction procedure

Fig. 3 illustrates the flowchart of our proposed method in this paper, including data acquisition, data preprocessing, model training and validation, and model evaluation. The steps of the RUL prediction process are summarized as follows:

Step 1: Collect battery capacity data under discharge conditions.

Step 2: Clean capacity data by removing outliers, and then divide the cleansed dataset into training and testing sets, followed by min-max normalization, with appropriate prediction starting points established.

Step 3: Input the preprocessed data into the model for training to obtain the nonlinear mapping relationship between historical capacities and future capacity. The first 80 % of the data in the training set is used for model training, while the remaining 20 % is utilized for model validation.

Step 4: Evaluate and visualize the RUL prediction results on three public datasets. Five performance metrics are used to evaluate the performance of the model under different prediction starting points, and the corresponding average metrics are calculated.

4. Experiments

4.1. Hyperparameter settings and evaluation metrics

Hyperparameter optimization is a pivotal element influencing the performance of lithium-ion battery RUL prediction models. Adaptive optimization algorithms can attain the best configuration of network hyperparameters, thereby making the network architecture more rational and efficient. These algorithms also circumvent the problem of the model being trapped in local optima due to manual parameter tuning. In this study, we opted for the Tree-structured Parzen Estimator (TPE) [49], a probabilistic model-based approach provided by the Optuna library, to perform hyperparameter optimization for our proposed PatchFormer model. The ranges for key hyperparameters, such as

the learning rate and dropout rate, along with the settings of those hyperparameters that were not adjusted, are outlined in Table 4. Furthermore, the number of trials was fixed at 200, and the Adam optimizer was utilized. All experiments were performed on a workstation with a Dual Intel (R) Xeon (R) E5-2643 v4 CPU @3.4 GHz with 12 cores and 8 NVIDIA TESLA V100 GPUs. All models used in this paper were built using the PyTorch framework and the pytorch_forecasting library. Each model repeated the experiment 10 times with different random seeds under the same set of hyperparameters and averaged. An early stopping mechanism was introduced to speed up the training process.

Assuming an EOL of $C_0 \times 70\%$ (C_0 represents the rated capacity of the battery), the actual and predicted RUL values of the lithium-ion batteries can be described as:

$$TRUL = T_{EOL} - T_{SP} \quad (18)$$

$$PRUL = \widehat{T}_{EOL} - T_{SP} \quad (19)$$

where T_{EOL} is the number of cycles at which the battery's life ends, $\widehat{T}_{EOL}$ is the predicted number of cycles at which the battery's life ends, and T_{SP} is the number of cycles already used by the battery at the starting point (SP) of the prediction process. In this study, for each prediction SP, we use five evaluation metrics to measure the model's prediction performance: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R-squared (R^2), Absolute Error (AE), and Relative Error (RE). The calculation formulas for these evaluation metrics are as follows:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (20)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (21)$$

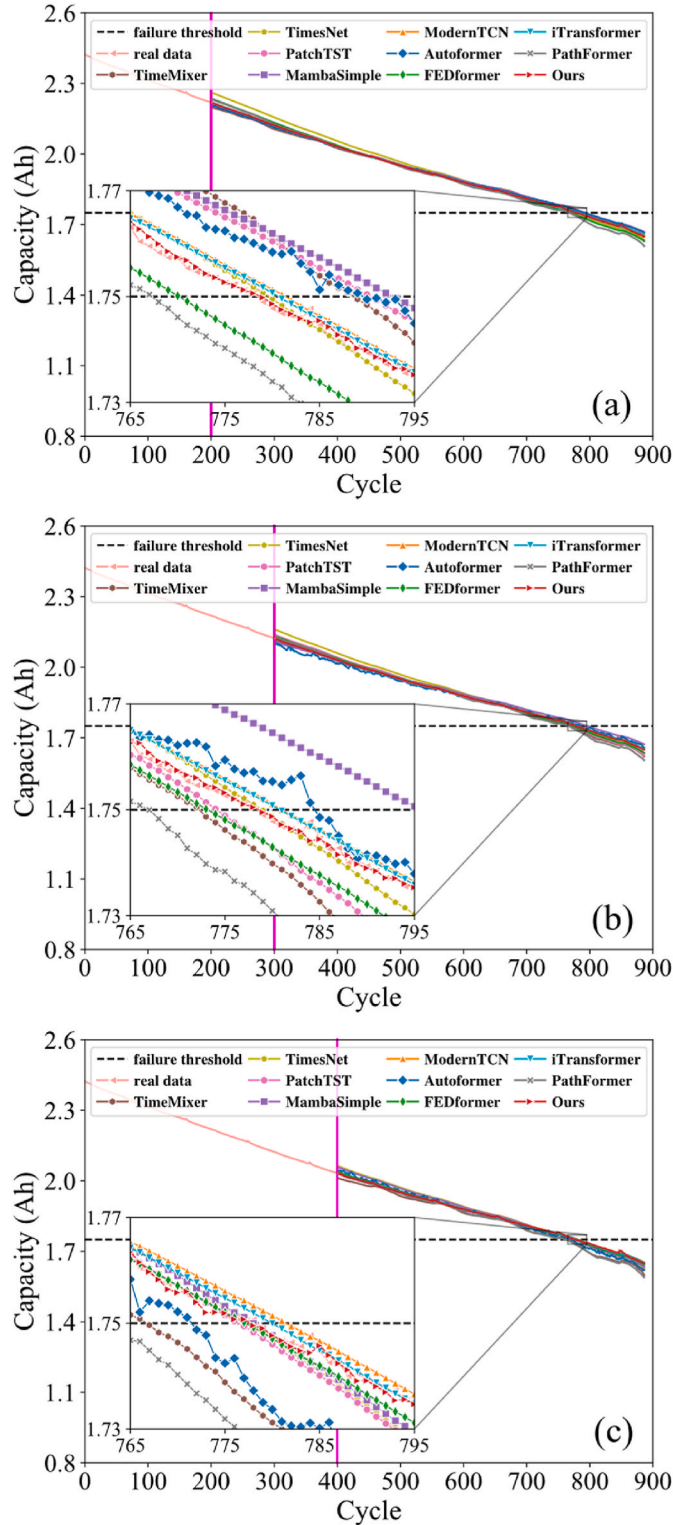


Fig. 7. Comparison of RUL prediction curves from different methods for TJU dataset. (a) prediction curves for SP = 200; (b) prediction curves for SP = 300; (c) prediction curves for SP = 400.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (22)$$

$$AE = |TRUL - PRUL| \quad (23)$$

$$RE = \frac{|TRUL - PRUL|}{TRUL} \quad (24)$$

where y_i represents the actual battery capacity value, $\bar{y}$ represents the mean of the actual battery capacity values, $\hat{y}_i$ represents the predicted value, $PRUL$ denotes the predicted value of RUL, and $TRUL$ denotes the true value of RUL. For the metrics MAE , $RMSE$, AE , and RE , the higher the RUL prediction accuracy is obtained when they are closer to zero. For R^2 , the higher value indicates the more accurate RUL prediction results.

To distill a more holistic assessment, we also average the aforementioned metrics across three different prediction SPs to derive five additional evaluation metrics: Average Mean Absolute Error ($AMAE$), Average Root Mean Squared Error ($ARMSE$), Average R-squared (AR^2), Average Absolute Error (AAE), and Average Relative Error (ARE).

4.2. Performance comparison for RUL prediction

To better evaluate the PatchFormer model, we employed nine classical TSF models—TimeMixer [50], TimesNet [51], PatchTST [52], MambaSimple [53], ModernTCN [54], Autoformer [30], FEDformer [31], iTransformer [55] and PathFormer [36]—as benchmarks for comparison. In order to comprehensively evaluate the RUL prediction performance of different models, RUL predictions were performed with the SPs of 300, 400 and 500 for CALCE I and CALCE II datasets, the SPs of 50, 70 and 90 for NASA dataset and the SPs of 200, 300 and 400 for TJU dataset.

4.2.1. CALCE I prediction results

After a detailed analysis of the RUL prediction results on the CALCE I dataset, as shown in Table 5, the following conclusions are drawn: a) Under various SP conditions, PatchFormer consistently demonstrates the lowest MAE, RMSE, AE, and RE metrics, as well as the highest R^2 value. In particular, the $TRUL$ value prediction error (AE) of PatchFormer is always within 1 cycle. In contrast, ModernTCN ranks second, while PathFormer and TimesNet occupy the lowest positions. For instance, when $SP = 300$, PatchFormer has an MAE of 0.0057, an RMSE of 0.0129, an R^2 of 0.9962, an AE of 0.8, and an RE of 0.0023. In comparison, PathFormer shows an MAE of 0.0431, an RMSE of 0.0640, an R^2 of 0.9060, an AE of 40.3, and an RE of 0.1185 under the same SP condition. b) The performance metrics of various models change under different SP conditions, indicating that the RUL prediction performance is influenced by the prediction SP. Evidently, the RUL prediction results of PatchFormer are less affected by changes in the prediction SP. c) PatchFormer outperforms all other models across the five statistical average metrics. d) Although the distributions of $AMAE$ s and $ARMSE$ s for the ten models are similar, the distributions of AAE s and ARE s are also similar but different from the former. This indicates that some models perform well in predicting the actual capacity degradation curve ($AMAE$ and $ARMSE$) but may perform poorly in predicting $TRUL$ values (AAE and ARE). For example, FEDformer shows better $AMAE$ (0.0227) and $ARMSE$ (0.0332) than Autoformer ($AMAE$: 0.0246, $ARMSE$: 0.0371), yet its AAE (16.0) and ARE (0.0781) are worse than those of Autoformer (AAE : 11.1, ARE : 0.0539).

The local magnified graphs shown in Fig. 4 indicate that under different SP conditions, PatchFormer's prediction curves are much closer to the actual capacity degradation curve compared to the other nine models. This further illustrates that PatchFormer provides the best prediction of the actual capacity degradation curve, aligning with the

Table 9

Ablation experiment of our proposed model on the CALCE I, CALCE II, NASA and TJU datasets. PA: Patch Attention; DPAN: Dual Patch-wise Attention Network.

Dataset	Method	SP	TRUL	PRUL	MAE	RMSE	R ²	AE	RE	AMAE	ARMSE	AR ²	AAE	ARE
CALCE I	Ours	300	341	340.4	0.0057	0.0129	0.9962	0.8	0.0023	0.0063	0.0141	0.9951	0.9	0.0045
		400	241	240.2	0.0062	0.0140	0.9954	0.8	0.0034					
		500	141	141.1	0.0069	0.0155	0.9937	1.1	0.0078					
	w/o Inter PA	300	341	340.7	0.0062	0.0130	0.9961	0.9	0.0026	0.0067	0.0143	0.9950	0.8	0.0039
		400	241	241.0	0.0066	0.0142	0.9953	0.8	0.0034					
		500	141	141.2	0.0072	0.0156	0.9936	0.8	0.0057					
	w/o Intra PA	300	341	341.2	0.0059	0.0130	0.9961	1.0	0.0029	0.0066	0.0143	0.9950	1.0	0.0051
		400	241	241.3	0.0065	0.0142	0.9953	0.9	0.0037					
		500	141	141.8	0.0073	0.0157	0.9935	1.2	0.0086					
	w/o DPAN	300	341	340.6	0.0061	0.0130	0.9961	1.0	0.0029	0.0068	0.0143	0.9949	1.0	0.0050
		400	241	240.6	0.0068	0.0142	0.9953	1.0	0.0042					
		500	141	141.3	0.0076	0.0158	0.9934	1.1	0.0079					
CALCE II	Ours	300	417	417.6	0.0047	0.0083	0.9984	1.0	0.0024	0.0051	0.0089	0.9981	1.0	0.0034
		400	317	317.8	0.0050	0.0088	0.9981	1.0	0.0032					
		500	217	217.8	0.0055	0.0096	0.9977	1.0	0.0046					
	w/o Inter PA	300	417	416.9	0.0051	0.0086	0.9982	1.7	0.0041	0.0055	0.0092	0.9979	1.5	0.0051
		400	317	317.3	0.0054	0.0092	0.9980	1.5	0.0047					
		500	217	217.8	0.0059	0.0099	0.9975	1.4	0.0065					
	w/o Intra PA	300	417	418.1	0.0048	0.0086	0.9983	1.1	0.0026	0.0052	0.0092	0.9980	1.2	0.0041
		400	317	318.2	0.0052	0.0091	0.9980	1.2	0.0038					
		500	217	218.3	0.0057	0.0099	0.9976	1.3	0.0060					
	w/o DPAN	300	417	418.3	0.0053	0.0089	0.9981	1.3	0.0031	0.0058	0.0096	0.9977	1.4	0.0047
		400	317	318.4	0.0057	0.0095	0.9978	1.4	0.0044					
		500	217	218.4	0.0064	0.0104	0.9973	1.4	0.0065					
NASA	Ours	50	75	75.1	0.0056	0.0118	0.9924	0.1	0.0014	0.0062	0.0128	0.9807	0.4	0.0088
		70	55	55.4	0.0061	0.0127	0.9835	0.4	0.0074					
		90	35	34.6	0.0068	0.0140	0.9663	0.6	0.0176					
	w/o Inter PA	50	75	74.0	0.0057	0.0119	0.9923	1.0	0.0135	0.0063	0.0132	0.9796	0.9	0.0203
		70	55	54.5	0.0063	0.0131	0.9825	0.5	0.0093					
		90	35	34.3	0.0069	0.0145	0.9639	1.3	0.0382					
	w/o Intra PA	50	75	74.9	0.0059	0.0119	0.9922	0.1	0.0014	0.0063	0.0130	0.9801	0.6	0.0153
		70	55	55.0	0.0061	0.0129	0.9831	0.5	0.0093					
		90	35	34.2	0.0069	0.0143	0.9651	1.2	0.0353					
	w/o DPAN	50	75	73.1	0.0062	0.0122	0.9918	2.1	0.0284	0.0064	0.0133	0.9793	1.7	0.0377
		70	55	54.4	0.0064	0.0132	0.9822	0.6	0.0111					
		90	35	34.0	0.0067	0.0145	0.9640	2.5	0.0735					
TJU	Ours	200	579	579.2	0.0013	0.0019	0.9999	0.8	0.0014	0.0014	0.0020	0.9997	0.9	0.0019
		300	479	478.9	0.0014	0.0020	0.9997	0.5	0.0011					
		400	379	378.6	0.0015	0.0022	0.9996	1.3	0.0032					
	w/o Inter PA	200	579	581.2	0.0015	0.0022	0.9998	2.2	0.0038	0.0016	0.0024	0.9996	2.2	0.0046
		300	479	481.3	0.0015	0.0023	0.9997	2.3	0.0048					
		400	379	381.0	0.0018	0.0026	0.9994	2.0	0.0053					
	w/o Intra PA	200	579	580.8	0.0017	0.0023	0.9998	1.8	0.0031	0.0017	0.0024	0.9996	2.0	0.0045
		300	479	481.0	0.0016	0.0024	0.9997	2.0	0.0042					
		400	379	381.3	0.0017	0.0025	0.9994	2.3	0.0061					
	w/o DPAN	200	579	580.1	0.0018	0.0025	0.9997	1.1	0.0019	0.0017	0.0025	0.9996	1.5	0.0033
		300	479	480.4	0.0017	0.0024	0.9997	1.4	0.0029					
		400	379	380.9	0.0017	0.0025	0.9994	1.9	0.0050					

highest R² value presented in Table 5. Additionally, PatchFormer excels not only in predicting the normal aging region of the capacity curve but also effectively captures and fits the phenomena of capacity regeneration.

Fig. 5 displays three scatter plots indicating that under the identical SP conditions, PatchFormer performs the worst in terms of MAE and RMSE, while PatchFormer's MAE and RMSE are closest to zero, indicating the best performance. ModernTCN's MAE and RMSE are slightly inferior to those of PatchFormer. Overall, the trend suggests that both MAE and RMSE increase with increasing SP values. The radar chart demonstrates that PatchFormer consistently outperforms other baseline models in terms of the R² metric across different SP conditions, while all models exhibit a declining trend in R² as SP increases. The two bar charts illustrate that PatchFormer consistently outperforms other baseline models in terms of AE and RE metrics.

4.2.2. CALCE II prediction results

We further evaluated the RUL prediction performance of the PatchFormer model on the CALCE II dataset. From Table 6, we can draw conclusions similar to those obtained from the CALCE I dataset. In the CALCE II dataset, all metrics for the PatchFormer model outperform

those of the other baseline models.

The local magnified views in Fig. S1 indicate that, under the identical SP conditions, the prediction curves of PatchFormer are much closer to the actual capacity degradation curve compared to the other nine models. This suggests that PatchFormer provides the best prediction for the actual capacity degradation curve, consistent with the highest R² value presented in Table 6. Besides, regarding the actual capacity degradation curve, PatchFormer not only provides highly accurate prediction in the normal aging region but also effectively captures and fits local capacity regeneration phenomena.

In Fig. S2, three scatter plots illustrate that under the identical SP conditions, TimesNet performs the worst in terms of MAE and RMSE, while PatchFormer's MAE and RMSE values are closest to zero, indicating its superior performance. The MAE and RMSE values for ModernTCN are slightly worse than those of PatchFormer. Overall, the trend indicates that both MAE and RMSE increase with higher SP values. The radar chart demonstrates that PatchFormer consistently outperforms baseline models in terms of the R² metric across different SP conditions. As SP values increase, the R² metrics for all models exhibit a downward trend. The two bar charts indicate that PatchFormer's AE and RE metrics are slightly higher than those of ModernTCN and iTransformer, but

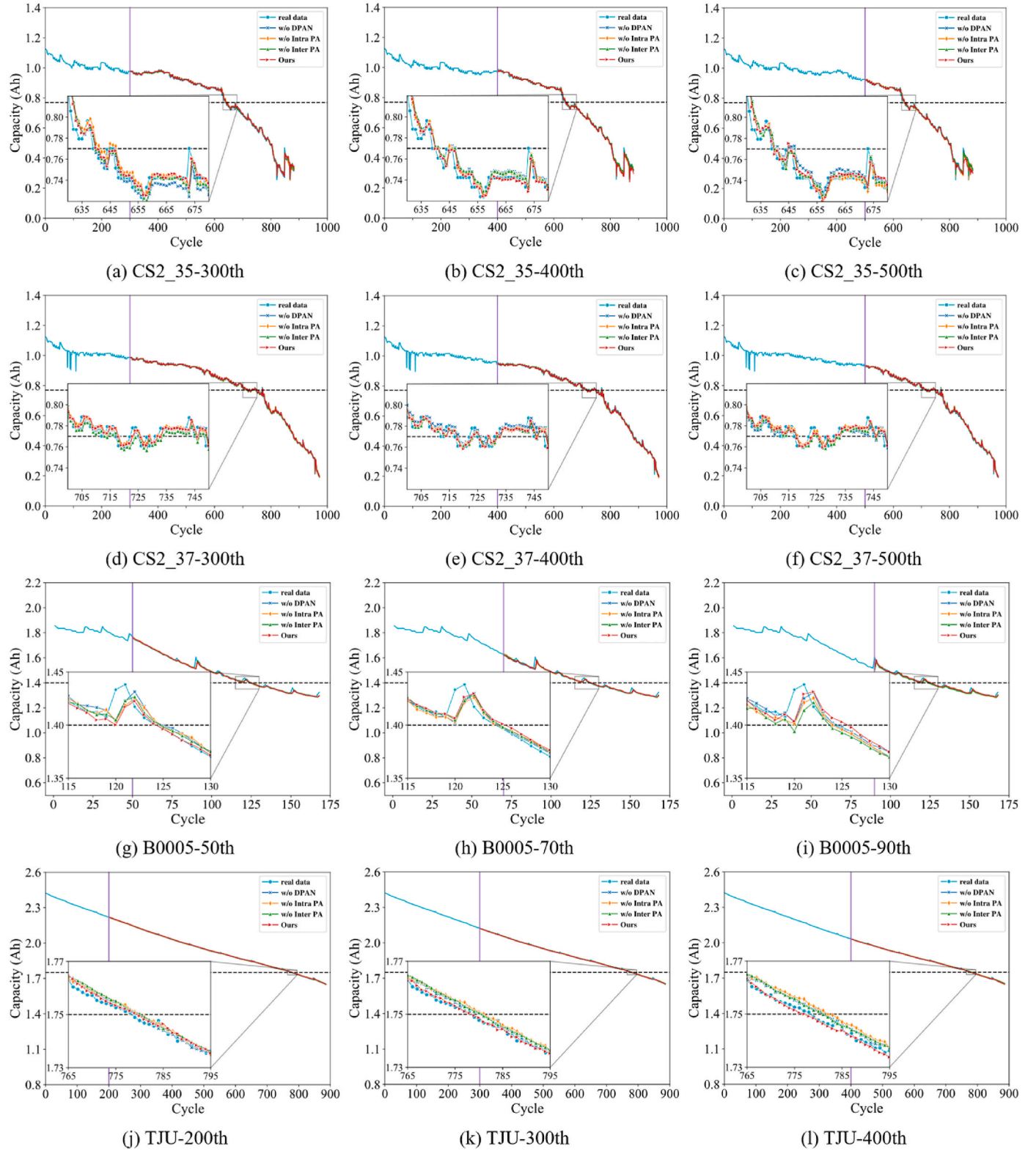


Fig. 8. Ablation experiment on the CALCE I, CALCE II, NASA and TJU datasets. (a) CS2_35-300th. (b) CS2_35-400th. (c) CS2_35-500th. (d) CS2_37-300th. (e) CS2_37-400th. (f) CS2_37-500th. (g) B0005-50th. (h) B0005-70th. (i) B0005-90th. (j) TJU-200th. (k) TJU-300th. (l) TJU-400th.

significantly lower than those of the other baseline models.

4.2.3. NASA prediction results

To further demonstrate the superiority of our proposed PatchFormer model, we conducted a more extensive study on the NASA dataset. A

comprehensive analysis of the RUL prediction results for the NASA dataset presented in Table 7 allows us to draw conclusions similar to those observed in the CALCE I and CALCE II datasets. In the NASA dataset, all metrics for the PatchFormer model outperform those of the other baseline models.

As shown in Fig. 6, the local magnified images at EOL indicate that, under the identical SP conditions, the prediction curves of PatchFormer are closer to the actual capacity degradation curve than those of the other nine models. This finding reinforces the assertion that PatchFormer provides the best prediction for the actual capacity degradation curve, as evidenced by the highest R^2 value in Table 7. Moreover, for the actual capacity degradation curve, PatchFormer not only delivers highly accurate prediction in the normal aging region but also effectively captures and models local capacity regeneration phenomena.

In Fig. S3, three scatter plots illustrate that PatchFormer's MAE and RMSE values are the closest to zero, indicating superior performance. The MAE and RMSE values for iTransformer are slightly worse than those for PatchFormer. Overall, this trend suggests that both MAE and RMSE decrease with increasing SP values. This conclusion contrasts with the findings from the CALCE dataset. The difference arises because this study did not include data prior to the prediction SP in the training set for the CALCE dataset, while for the NASA dataset, the data prior to the prediction SP was included as part of the training set. As SP values increase, the training data expands, enhancing the model's generalization ability for unknown data, which subsequently reduces the MAE and RMSE metrics. The radar chart indicates that PatchFormer consistently outperforms other baseline models in terms of the R^2 metric across different SP conditions. With the increase in SP, almost all models exhibit a downward trend in their R^2 metrics, except for TimesNet. The two bar charts demonstrate that PatchFormer achieves better AE and RE metrics compared to the other baseline models.

4.2.4. TJU prediction results

To further validate the generalization capability of our model, we conducted experiments using the NCM + NCA batteries CY25_1, CY25_2, and CY25_3 from the newer TJU dataset. Compared to the CALCE and NASA datasets, the capacity degradation data collected in the TJU dataset features smoother actual capacity degradation curves, with a lower frequency and smaller magnitude of capacity regeneration phenomena. As a result, the model finds it easier to predict the actual capacity degradation curve, leading to lower MAE and RMSE values, as well as a higher R^2 . As shown in Table 8 and Fig. S4, our proposed PatchFormer model outperforms the majority of evaluation metrics, only slightly trailing behind the TimesNet model in the AE and RE metrics. However, TimesNet ranks lowest among the ten models in other metrics, demonstrating significantly inferior overall performance compared to PatchFormer.

Fig. 7 illustrates that under different SP conditions, the prediction results of the PatchFormer model have the best fitting effect on the actual capacity degradation curve. It effectively predicts the normal aging region of the capacity degradation curve while adequately fitting the capacity regeneration region. The newer TJU dataset better reflects the latest battery technologies and usage patterns, providing a solid foundation for evaluating the performance of the proposed model and its relevance to contemporary applications. The experimental results on the TJU dataset further validate the effectiveness and robustness of the PatchFormer model, offering strong support for its application in real-world scenarios involving newer battery systems.

4.3. Ablation experiments of network structure

To validate the effectiveness of our experimental design, ablation experiments were conducted on the CALCE I, CALCE II, NASA and TJU datasets, focusing on the impact of inter-patch attention and intra-patch attention on predictive performance. As shown in Table 9, removing inter-patch attention and intra-patch attention can lead to a decline in the model's prediction performance. Meanwhile, the elimination of dual patch-wise attention will further aggravate this decline. This confirms the effectiveness and necessity of the proposed model design. Specifically, in the CALCE II dataset, the presence of DPAN increases the model's performance metrics AMAE, ARSME, AAE, and ARE by 0.07 %,

0.07 %, 0.4 %, and 0.13 %, respectively, while AR^2 decreases by 0.04 %. This is attributed to the DPAN's utilization of a dual attention mechanism, which effectively extracts and integrates both local details and global correlations of time series information. Additionally, Table 9 indicates that removing the inter-patch attention results in a greater decline in model performance compared to the removal of the intra-patch attention. This suggests that extracting global correlations from time series information is more effective in enhancing model prediction performance than extracting local details.

Fig. 8 illustrates that, under various SP conditions, the capacity prediction curves of the PatchFormer model closely align with the actual capacity degradation curves in both the CALCE I, CALCE II, NASA and TJU datasets. Compared to other ablation models, PatchFormer demonstrates superior predictive capability, achieving optimal performance in the normal aging region of the battery and effectively capturing the capacity regeneration phenomena. This indicates that both the inter-patch and intra-patch attention modules in the PatchFormer model significantly contribute to its performance improvement.

5. Conclusion

To enhance the accuracy of RUL prediction for lithium-ion batteries, this paper proposes a patch-based transformer (PatchFormer). The designed DPAN can effectively extract and fuse the global and local features of the partitioned time series. The constructed FAN uses self-attention on the fused time features to capture the correlation between features. PatchFormer can not only effectively characterize the degradation trend in the battery capacity sequence, but also accurately capture the capacity regeneration phenomena of lithium-ion batteries. In addition, the model's performance is optimized by employing the TPE algorithm to fine-tune the network's hyperparameters.

To evaluate the performance of PatchFormer, we benchmark it against various baseline models, specifically TimeMixer, TimesNet, PatchTST, MambaSimple, ModernTCN, Autoformer, FEDformer, iTransformer, and PathFormer, which are nine TSF models. The capabilities of PatchFormer are demonstrated across multiple publicly available datasets, with the following conclusions:

- (1) On the CALCE I, CALCE II and NASA datasets, PatchFormer achieves the minimum values for MAE, RMSE, AE, and RE, along with the maximum R^2 values, under different SP conditions. Moreover, the related statistical average metrics AMAE, ARMSE, AR^2 , AAE, and ARE are also optimal. On the TJU dataset, most of the metrics of PatchFormer are also the best. This shows that compared with other models, PatchFormer demonstrates superior prediction performance both in forecasting the true capacity degradation curve and in predicting the true value of RUL, and it possesses advantages such as being less influenced by the prediction SP and having strong generalization capability.
- (2) Ablation experiments demonstrate that dual patch-wise attention improves the RUL prediction performance, verifying the effectiveness and necessity of the network structure design. This is because dual patch-wise attention can model both the local connections within the individual patch and the global connections between divided patches to fully extract the dependencies between temporal information of partitioning time series.

Overall, the proposed PatchFormer model in this paper not only achieves reliable RUL prediction results but also accurately predicts the capacity regeneration and degradation trends of the battery capacity sequence. Future work will further explore how to incorporate various factors that affect battery behavior (including temperature and current) into the model. In terms of model framework design, we will consider combining data-driven models with lithium-ion battery physicochemical models.

CRedit authorship contribution statement

Lei Liu: Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization. **Jiahui Huang:** Writing – original draft, Visualization, Software, Methodology, Data curation, Conceptualization. **Hongwei Zhao:** Writing – review & editing, Validation, Investigation. **Tianqi Li:** Validation, Supervision. **Bin Li:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research is supported by the Anhui Provincial Natural Science Foundation (Grant No.2408085QF214), the Fundamental Research Funds for the Central Universities (Grant No. WK2100000045), and the Opening Project of the State Key Laboratory of General Artificial Intelligence (Grant No. SKLAGI2024OP10, Grant No. SKLAGI2024OP11).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2025.236187>.

Data availability

Data is openly accessible and available for utilization.

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